

Role Specialization and Context Fragmentation in Multi-Agent LLM Systems: An Empirical Study on Procurement Automation

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Abstract

This paper investigates under what conditions multi-agent specialization outperforms a single-agent approach in procurement automation, and when the division into specialized roles generates context fragmentation that prevents systemic risk detection. In the architectures we evaluate, each agent perceives only a portion of the problem — no single agent holds a complete representation. Robotic Process Automation (RPA) systems handle routine tasks well but fail on exceptions requiring reasoning, such as contracts containing exclusivity clauses. Multi-agent systems based on large language models (LLMs) emerge as a promising alternative in that setting.

We propose a framework of five specialized agents orchestrated with CrewAI. Nine variants were evaluated across 50 synthetic cases and 50 real cases drawn from the BPI Challenge 2019 [1]. The Multi-Mixed variant (8B agents with a 70B decision-maker) achieved the highest Autonomous Resolution Rate (ARR = 98.7%). The results provide empirical evidence consistent with the compute equivalence hypothesis: a single agent with extended context can match or exceed multi-agent performance under the same token budget. That said, HDA — an exploratory metric given the small number of positive cases ($n = 13$) — suggests that the most autonomous configurations may have limitations in detecting risks that require human intervention, though these differences lack statistical power to be conclusive.

Our primary contribution is an empirical analysis of the trade-offs between autonomy and systemic risk detection under a controlled token budget, including a role-level ablation study and validation on real BPI Challenge 2019 data. The findings suggest that reliable risk detection will likely require

redesigning the information flows between agents to mitigate context fragmentation — rather than simply adding more agents or scaling up model size.

Keywords: Multi-agent systems, Large Language Models, Procurement automation, Human-in-the-Loop, Compute equivalence, Context fragmentation, BPI Challenge 2019

1. Introduction

A technician needs to purchase a spare part. The supplier is approved, budget is available, and the urgency is real. Yet the request will take between three and seven days to receive a response, because it must pass sequentially through the procurement, legal, finance, and compliance departments. The administrative cost of processing that order is not only financial — it also includes the time burden placed on staff.

The problem is not bureaucracy itself, but that the systems designed to automate it lack any capacity for reasoning. Robotic Process Automation (RPA) systems execute repetitive, well-structured tasks efficiently. They fail, however, when faced with regulatory deviations or conflicts of interest. These systems do not reason, nor do they escalate appropriately. They simply stop and wait for human intervention [2].

Multi-agent frameworks based on Large Language Models (LLMs) can go beyond fixed rules: they can reason in the face of the unexpected. Tools such as AutoGen [3], CrewAI [4], and LangGraph [5] make it possible to build teams of specialized agents capable of collaborating on exceptions. Work such as ChatDev [6] showed that these teams can outperform a single agent on complex tasks, while MetaGPT [7] and AgentVerse [8] extended those results through structured roles and inter-role coordination.

A question the literature has not yet answered well remains: when is it actually worth using five agents instead of one? And when does specialization become part of the problem rather than the solution? Evidence in real enterprise environments is still sparse [9]. This work attempts to address that gap.

The main contributions of this study are: (i) a replicable multi-agent framework across five business processes, together with evaluation data and published source code; (ii) an empirical analysis of the architectural trade-offs among *compute equivalence*, model diversity, and systemic risk detection;

and (iii) a risk signal taxonomy validated by six independent annotators ($\kappa=0.652$), which helps explain why detection capacity varies substantially across domains.

The results suggest that multi-agent specialization introduces a trade-off between autonomy and risk detection that cannot be resolved simply by adding more agents or scaling up model size.

Research question. Under what conditions does multi-agent specialization outperform a single-agent approach in procurement process automation, and when does context fragmentation impair systemic risk detection? Do these patterns replicate on real-world data from the BPI Challenge 2019 [1]?

The paper is organized as follows: related work (Sec. 2), framework (Sec. 3), experimental setup (Sec. 4), results (Sec. 5), comparative analysis (Sec. 6), ablation study (Sec. 7), cross-process and BPI validation (Sec. 8), discussion (Sec. 9), and conclusions (Sec. 10). The nine evaluated variants are described in detail in Table 3 (Sec. 3).

2. Related Work

2.1. Multi-Agent LLM Systems

AutoGen [3] introduced a multi-agent conversation framework in which LLMs collaborate through structured messages. CrewAI [4] extends this paradigm with a role-and-task model inspired by human organizations. Talebirad et al. [10] analyzed multi-agent collaboration as a mechanism for amplifying the collective capacity of LLMs [9]. Park et al. [11] documented emergent social behaviors in agent communities. ChatDev [6] showed that specialized agents can outperform a single agent on software development tasks. MetaGPT [7] assigned software engineering roles to different agents and found that specialization reduces hallucinations in code-related tasks. CAMEL [12] proposed role-driven communication as a coordination mechanism — a direct antecedent of the design used here. AgentVerse [8] found that team composition affects performance in a non-linear fashion. Li et al. [13] showed that scaling the number of agents improves performance when the base model has sufficient capacity — a result consistent with the 8B–70B threshold identified in this work.

Modern LLMs combine chain-of-thought reasoning [14] and reactive acting [15], which are the underlying cognitive mechanisms of the specialized agents we evaluate. Factual hallucinations [16] represent the primary risk

in enterprise settings, which motivates the Human-in-the-Loop (HiL) mechanism introduced in this work.

2.2. Business Process Automation

RPA systems — UiPath [17] and Automation Anywhere [18] — automate repetitive tasks well, but their limitation is well documented: when an unforeseen exception appears, they stop [2]. That is precisely what motivates this work.

Benchmarks such as WorkArena [19] and TaskBench [20] exist for evaluating agents in enterprise environments, though neither targets procurement specifically with annotated ground truth. ToolBench [21] and Toolformer [22] showed that LLMs can learn to use real APIs autonomously, which supports the role-specific tool design used here. APIBench [23] goes a step further by evaluating whether LLMs generate correct API calls — the same type of validation we apply to the agents in this study.

Process mining [24, 25] provides the methodological framework for analyzing processes from event logs. The BPI Challenge 2019 [1, 26] — a real-world Purchase-to-Pay event log with 251,734 cases — serves as the external validation benchmark for this work and as the source for parametrizing the APIs with real statistics. The GDPR framework [27] defines the regulatory requirements the Compliance agent must verify, and the Human-in-the-Loop mechanism [28] ensures human oversight on the highest-risk decisions.

2.3. Compute Equivalence and Model Diversity

Wu et al. [3] argue that fair comparisons across architectures should control for token budget, not just agent count. We carry out that comparison empirically across three configurations: Single-Small (200 tokens), Single-Small-XL (750 tokens), and Multi-Small (750 tokens, 5 agents). If a single agent with more context matches the multi-agent system, the architecture is not adding real value. The experimental details are in Section 6.

The experiments use two models from the Llama 3 family [29]: `llama-3.1-8b-instant` for the auxiliary agents and `llama-3.3-70b-versatile` as the decision-maker in the Mixed and Large variants. Both run via Groq [30], which enables 70B-model inference with latencies below 3 seconds per case. GPT-3 [31] and GPT-4 [32] are included as state-of-the-art reasoning references, though they are not used directly in the experiments.

In ensemble learning, mixing models of different sizes reduces error correlation [33, 34]. That principle is what we test by comparing Multi-Small ($8B \times 5$), Multi-Mixed ($8B \times 4 + 70B \times 1$), and Multi-Large ($70B \times 5$). The question is whether model diversity matters more than architecture. Benchmarks such as SWE-bench [35] and GAIA [36] have set the standard for evaluating agents on real tasks; we adapt that methodology here to the procurement domain with our own ground truth.

Table 1 positions this work relative to the most relevant prior studies.

Table 1: Comparison with related work. None of the prior studies evaluates the trade-off between multi-agent specialization and systemic risk detection on real process mining data.

Work	Architecture	Dataset	HDA	Gap
AutoGen [3]	Multi-agent	Synthetic	Not eval.	No risk detection
ChatDev [6]	Multi-agent	SWE-bench	Not eval.	SW development only
AgentVerse [8]	Multi-agent	204 tasks	Not eval.	No real enterprise data
WorkArena [19]	Single-agent	ServiceNow	Not eval.	No HiL mechanism
This work	Multi + Single	50 synth. + BPI 2019	Evaluated	Proc. + HiL + real data

3. Proposed Framework

3.1. General Architecture

The system is structured as a linear Directed Acyclic Graph (DAG) that reflects how a procurement process actually works: budget is validated first, then the contract is reviewed, and only at the end is the compliance decision made. This ordering is not arbitrary — it follows real process dependencies. The orchestrator is CrewAI. In HiL variants, the Compliance agent has explicit escalation rules that trigger whenever it detects critical risk.

A graph with cycles would introduce circular dependencies that complicate implementation without clear benefit in this domain (see Table 2 for a comparison of alternatives). This architectural constraint — the fixed DAG with no iteration — is documented in the Limitations section (Sec. 9.3) and motivates the cross-process synthesis hypothesis.

Table 2: Comparison of orchestration topologies for multi-agent procurement systems.

Topology	Advantages	Limitations	Applicability
Linear DAG (CrewAI)	Simple, predictable, easy to debug	Context fragmentation, no iteration	✓ Natural procurement flow
Graph with cycles	Iterative feedback	Loop risk, higher latency	× Circular dependencies
Parallel processing	Time efficiency	Synchronization complexity	~ Partially applicable
Adaptive graphs (LangGraph)	Dynamic conditional flows	Higher implementation complexity	~ Future work

Analyzing the topology as an inter-agent communication network reveals a notable bottleneck: the Finance agent acts as a structural bridge between operational validation and regulatory decision-making. Its betweenness centrality score is the highest in the system, meaning that any information flowing from Procurement toward Legal passes necessarily through it.

The five agents and their main APIs are:

- **Requester:** registers the request via POST `/api/requests/new`
- **Procurement:** validates the supplier via GET `/api/suppliers/{id}`
- **Finance:** verifies budget via GET `/api/budget/{dept}`
- **Legal:** analyzes contractual clauses via POST `/api/contracts/validate`
- **Compliance:** final decision-maker; escalates to a human via POST `/api/escalation/human` when critical risk is detected; queries the GDPR/CE/PRL/AESA regulatory database via GET `/api/regulations/{cat}`

The complete system prompts for each agent and the endpoint table are included in Appendix A and in the public repository.

3.2. The Nine System Variants

Table 3 summarizes the configurations. Variants marked with * report mean $\pm \sigma$ across three independent replications; the rest are single runs.

Table 3: Summary of variants and metrics. ARR: Autonomous Resolution Rate; HDA: HiL Detection Accuracy; DER: Decision Error Rate; PT: processing time per case (s). Variants marked * report mean $\pm \sigma$ (3 replications). Justification quality (SAT) reported in Appendix E.

Model	Config.	ARR	HDA	DER	PT	Purpose
RPA-Baseline*	if-then rules	100% $\pm$ 0.0	0% $\pm$ 0.0	34% $\pm$ 0.0	$\approx$ 0	Minimum baseline
Single-Small*	8B $\times$ 1	92% $\pm$ 11.3	10.3% $\pm$ 14.5	27.3% $\pm$ 3.8	0.31	LLM reference
Single-Small-XL*	8B $\times$ 1 (750 tok)	79.0% $\pm$ 3.2	36.5% $\pm$ 13.1	20.5% $\pm$ 5.7	0.40	Compute eq.
Multi-Small*	8B $\times$ 5	79.5% $\pm$ 7.1	21.2% $\pm$ 10.6	27.8% $\pm$ 4.7	1.74	Main multi
Multi-Small-Abl	8B $\times$ 3	88%	0%	30%	0.77	Ablation
Multi-Mixed*	8B $\times$ 4+70B $\times$ 1	98.7% $\pm$ 0.9	5.1% $\pm$ 3.6	27.3% $\pm$ 0.9	1.69	Best ARR/cost
Multi-Large	70B $\times$ 5	98%	0%	26%	2.40	Homog. powerful
Multi-Mixed-HiL	8B $\times$ 4+70B $\times$ 1+HiL	98%	0%	28%	1.56	Active HiL
Multi-Mixed-HiL-Clf	2-stage, 8B	94%	23.1%	22%	2.85	Best dep. HDA
<i>AutoGen 0.7.5</i>	<i>RoundRobin, 8B$\times$5</i>	<i>100%</i>	<i>0%</i>	<i>76%</i>	<i>6.55</i>	<i>Alt. baseline</i>

4. Experimental Setup

4.1. Dataset and Metrics

We designed 50 procurement cases with annotated ground truth at three levels: *Simple* (C01–C20, standard purchases), *Medium* (C21–C38, operational exceptions), and *Complex* (C39–C50, regulatory and strategic exceptions).

Of the 50 cases, 13 require HiL escalation. The distribution follows the methodology of WorkArena [19] and AgentBench [37] for controlled environments with annotated ground truth. Evaluation on real data was carried out on the BPI Challenge 2019 [1] (Sec. 8).

The main metrics are ARR (*Autonomous Resolution Rate*, target $> 70\%$, consistent with WorkArena [19] benchmarks for enterprise task automation), HDA (*HiL Detection Accuracy*, reported as an exploratory metric given the small number of positive cases, $n = 13$; target $> 90\%$ reflects the safety requirement that critical risk cases should rarely go undetected in regulated procurement), DER (*Decision Error Rate*, target $< 10\%$, aligned with acceptable error rates in automated business process systems [2]), and PT (*Process Time*). SAT (*Satisfaction Score*) is evaluated by LLM-as-Judge [38] on clarity, completeness, and actionability (1–5 scale, target > 3.5); note that this metric has not been validated against human evaluators and should be treated as indicative only.

4.2. Technology Stack and Reproducibility

- Framework: CrewAI, Python 3.12, Windows 10, Jupyter Lab

- Auxiliary LLM: `llama-3.1-8b-instant` via Groq API [30]
- Decision LLM (Mixed/Large): `llama-3.3-70b-versatile` via Groq API
- Temperature: 0.0 (with the caveat documented below)
- Anti-rate-limit pause: 4–8 s between cases depending on model
- Code: <https://github.com/Edurne0781/multiagent-procurement-llm>

A clarification on determinism is warranted. The `temperature=0.0` setting was verified in a single session: five test cases produced identical outputs. The robustness analysis across three independent replications (Table 4), however, shows σ ARR = 11.3 pp for the 8B model — consistent with the literature on non-determinism in distributed LLM inference [39]. Groq introduces variability even at fixed temperature due to differences in server state and dynamic batching. Multi-Mixed ($\sigma = 0.9$ pp) is considerably more stable and offers the best ARR/stability balance among the evaluated configurations, though its exploratory HDA limits its applicability in settings with strict risk detection requirements.

5. Results

5.1. Overview

Figure 1 shows the three main metrics. Table 3 reports the complete values with standard deviations.

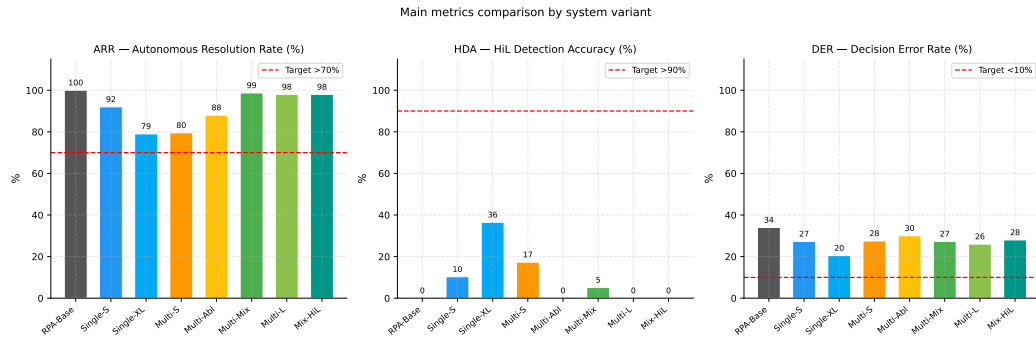


Figure 1: Comparison of ARR, HDA, and DER. Left: Autonomous Resolution Rate (target > 70%). Center: HiL Detection Accuracy (exploratory metric, $n = 13$ positive cases). Right: Decision Error Rate (target < 10%).

The most notable findings: Multi-Mixed achieves the highest ARR (98.7%) with minimal variability, but its HDA is very low (5.1%). Single-Small, using a smaller model, reaches $\text{HDA} = 10.3\%$ on average (30.8% in the first replication). Single-Small-XL and Multi-Small converge to nearly identical ARR means (79.0% vs. 79.5%) over four replications, consistent with the compute equivalence hypothesis. The RPA-Baseline delivers exactly what one would expect: $\text{ARR} = 100\%$ because it never escalates, $\text{HDA} = 0\%$, $\text{DER} = 34\%$.

Table 4: Results by replication. The variability of Single-Small ($\sigma \text{ ARR} = 11.3 \text{ pp}$) contrasts with the stability of Multi-Mixed when the decision-maker is the 70B model ($\sigma = 0.9 \text{ pp}$). Single-Small-XL and Multi-Small were run over four replications to verify the compute equivalence hypothesis under the high variance of the 8B model; the remaining variants were replicated three times. — indicates no fourth replication was conducted.

Model	Rep. 1	Rep. 2	Rep. 3	Rep. 4	Mean $\pm \sigma$
RPA-Baseline ARR	100%	100%	100%	—	100% $\pm 0.0 \text{ pp}$
RPA-Baseline HDA	0%	0%	0%	—	0% $\pm 0.0 \text{ pp}$
Single-Small ARR	76%	100%	100%	—	92% $\pm 11.3 \text{ pp}$
Single-Small HDA	30.8%	0%	0%	—	10.3% $\pm 14.5 \text{ pp}$
Single-Small DER	22%	28%	32%	—	27.3% $\pm 3.8 \text{ pp}$
Single-Small-XL ARR	84%	76%	76%	80%	79.0% $\pm 3.2 \text{ pp}$
Single-Small-XL HDA	23.1%	53.8%	30.8%	53.8%	40.4% $\pm 13.1 \text{ pp}$
Single-Small-XL DER	28%	24%	14%	16%	20.5% $\pm 5.7 \text{ pp}$
Multi-Small ARR	68%	82%	80%	88%	79.5% $\pm 7.1 \text{ pp}$
Multi-Small HDA	30.8%	7.7%	15.4%	15.4%	17.3% $\pm 8.5 \text{ pp}$
Multi-Small DER	28%	30%	26%	26%	27.5% $\pm 1.6 \text{ pp}$
Multi-Mixed ARR	100%	98%	98%	—	98.7% $\pm 0.9 \text{ pp}$
Multi-Mixed HDA	0%	7.7%	7.7%	—	5.1% $\pm 3.6 \text{ pp}$
Multi-Mixed DER	26%	28%	28%	—	27.3% $\pm 0.9 \text{ pp}$

The table shows that the ARR difference between Single-Small and Multi-Mixed is not statistically distinguishable when comparing replication means — the 6.7 pp gap falls within Single-Small’s standard error. The case-level McNemar test (Table 9) does reach significance, but addresses a different question: item-level agreement, not architectural stability. What the replication analysis makes clear is the stability difference: $\sigma = 0.9 \text{ pp}$ vs. 11.3 pp .

Single-Small-XL and Multi-Small were run over four replications to verify the compute equivalence hypothesis under the high variance of the 8B model ($\sigma = 11.3$ pp). Their means converge to nearly identical values ($79.0\% \pm 3.2$ pp vs. $79.5\% \pm 7.1$ pp), which confirms that with `llama-3.1-8b-instant` neither expanding the context window nor adding specialized agents produces a consistent advantage. The remaining variants were replicated three times, consistent with standard practice for non-deterministic LLM evaluation [39].

Table 5: Confusion matrix for the 13 HiL cases ($n = 13$, `requires_hil=True`). TP: true positive (correctly escalated); TN: true negative; FP: false alarm; FN: undetected risk case.

Variant	TP	TN	FP	FN
Single-Small	4	29	8	9
Multi-Mixed	0	37	0	13
Multi-Mixed-HiL-Clf	3	37	0	10

Multi-Mixed produces no false alarms ($FP = 0$) but also detects no real risk cases ($FN = 13$). Multi-Mixed-HiL-Clf is the only deployable variant with perfect precision on the cases it escalates (0 false positives), though its recall remains low (3 out of 13).

5.2. Per-Variant Analysis

Note: values in this subsection correspond to the initial single run. Summary statistics (mean $\pm \sigma$ across 3 replications) are in Table 4.

RPA-Baseline. $ARR = 100\%$ because it never escalates — it approves or rejects mechanically. $HDA = 0\%$, $DER = 34\%$ (1 in every 3 decisions is wrong). In case C24, the RPA approves a contract containing an exclusivity clause because the supplier is registered and budget is available, without evaluating the legal risk. This is the lower bound of the experiment: any LLM-based system must improve on this DER.

Single-Small (8B, 200 tok). $ARR = 76\%$, $HDA = 30.8\%$, $DER = 22\%$. Most token- and time-efficient ($PT = 0.31$ s). Inter-replication variability is high ($\sigma = 11.3$ pp in ARR): in replications 2 and 3, HDA drops to 0% , which limits the interpretability of that value.

Multi-Small ($8B \times 5$). ARR = $79.5\% \pm 7.1$ pp (mean over 4 replications), virtually indistinguishable from Single-Small-XL ($79.0\% \pm 3.2$ pp) at the same token budget. The first replication gave 68%, but subsequent runs showed higher variance, consistent with the non-deterministic behaviour of `llama-3.1-8b-instant` on distributed inference. Multi-agent architecture provides no consistent advantage with small models.

Multi-Mixed ($8B \times 4 + 70B \times 1$). ARR = $98.7\% \pm 0.9$ pp, the largest jump in the experiment (+19.2 pp over Multi-Small mean). The Compliance decision-maker uses the 70B model. DER = 27.3%, estimated cost \$0.0065/case. This is the most stable configuration in the experiment.

Multi-Large ($70B \times 5$). ARR = 98%, virtually identical to Multi-Mixed despite using powerful models throughout. It costs $2.7\times$ more for comparable results, which is consistent with the bottleneck in Multi-Mixed having been the decision-maker rather than the auxiliary agents.

Multi-Mixed-HiL-Clf. ARR = 94%, HDA = 23.1%, DER = 22%, PT = 2.85 s. Best deployable variant in terms of HDA. The 3 escalated cases match the ground truth `requires_hil=True` exactly, with 0 false positives.

Table 6: 95% confidence intervals for ARR and HDA. The high variability of Single-Small means its ARR difference from Multi-Mixed is not statistically distinguishable, though Multi-Mixed is substantially more stable.

Model	ARR (%)	95% CI ARR	HDA (%)	95% CI HDA
RPA-Baseline	100.0 ± 0.0	[100.0, 100.0]	0.0 ± 0.0	[0.0, 0.0]
Single-Small	92.0 ± 11.3	[69.0, 115.0] [†]	10.3 ± 14.5	[0, 39.3]
Multi-Mixed	98.7 ± 0.9	[96.9, 100.5]	5.1 ± 3.6	[1.5, 8.7]

[†]CI truncated to valid range [0, 100].

5.3. Sensitivity Analysis and Confidence Intervals

The ARR difference between Single-Small and Multi-Mixed (6.7 pp) is smaller than the standard error of Single-Small (11.3 pp), so it is not statistically significant in ARR terms. What differs substantially is stability: $\sigma = 0.9$ pp vs. 11.3 pp.

5.4. Computational Cost and Efficiency Frontier

Table 7: Token consumption and estimated cost per model. Costs are estimated at GPT-3.5-turbo market prices for comparability; in production with free-tier Groq, the actual cost is \$0.00.

Model	Tok/case	Total tok	Calls	Cost (\$)	Note
RPA-Baseline	0	0	0	\$0.0000	No LLM
Single-Small	200	10,000	1	\$0.0012	Most token-efficient
Single-Small-XL	750	37,500	1	\$0.0045	Compute eq. confirmed (4 reps.)
Multi-Small	750	37,500	5	\$0.0045	Indistinguishable from XL
Multi-Small-Abl	450	22,500	3	\$0.0027	40% saving vs. Multi-Small
Multi-Mixed	750	37,500	5	\$0.0065	Optimal cost/performance
Multi-Large	750	37,500	5	\$0.0120	2.7× Multi-Mixed, same ARR
Multi-Mixed-HiL	875	43,750	5	\$0.0053	Balanced ARR/cost

5.5. Justification Quality: LLM-as-Judge

Multi-Mixed produces the highest-quality justifications ($\text{SAT} = 4.36$) and the most consistent ones ($\sigma = 0.27$, the lowest across all variants). RPA-Baseline falls below the minimum threshold: justifications of the type “Supplier OK + budget OK” give the recipient no guidance on what to do (actionability = 2.40).

Full results are reported in Appendix E.

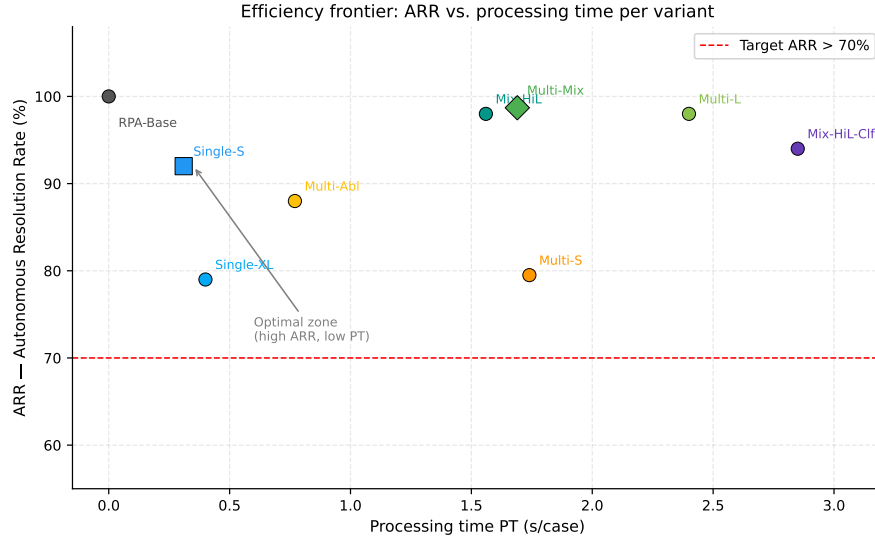


Figure 2: Efficiency frontier: ARR vs. processing time per case. Single-Small offers the best ARR/efficiency balance for lightweight deployments; Multi-Mixed reaches optimal ARR at moderate processing time.

Single-Small stands out as the most efficient option for lightweight deployments: it resolves 76% of cases in 0.31 seconds per case. Multi-Mixed reaches the highest ARR but requires 1.69 seconds — a reasonable time for enterprise settings where accuracy matters more than speed.

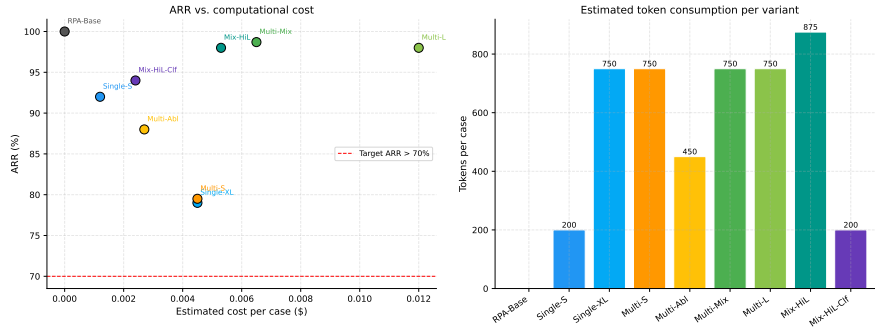


Figure 3: ARR vs. token cost (left) and estimated token consumption per model (right). Multi-Mixed is the optimal cost/performance point.

Table 8: McNemar test with continuity correction on the ARR criterion ($n = 37$ non-HiL cases). Significance: * $p < 0.05$, ** $p < 0.01$, n.s. not significant.

Comparison	ARR A	ARR B	b	c	Sig.
RPA-Baseline vs. Single-Small	100.0%	78.4%	8	0	*
Single-Small vs. Multi-Mixed	78.4%	100.0%	0	8	*
Single-Small-XL vs. Multi-Small	86.5%	83.8%	6	5	n.s.
Multi-Small vs. Multi-Mixed	83.8%	100.0%	0	6	*
Multi-Mixed vs. Multi-Large	100.0%	97.3%	1	0	n.s.

The test confirms that the most important differences are statistically significant, with one notable exception: Single-Small-XL vs. Multi-Small does not reach significance — exactly the result one would expect if the compute equivalence hypothesis holds.

5.6. Statistical Significance: McNemar Test

ARR differences are significant at the key steps: from RPA to Single-Small, from Single-Small to Multi-Mixed, and from Multi-Small to Multi-Mixed (all $p < 0.05$). The difference between Single-Small-XL and Multi-Small does not reach significance (n.s.), which is the expected result under the compute equivalence hypothesis.

For HDA ($n = 13$ HiL cases), no comparison reaches statistical significance, reflecting insufficient statistical power rather than true equivalence. The Bayes factor $BF_{10} = 0.41$ indicates anecdotal evidence in favor of H_0 for the HDA difference.

6. Comparative Analysis: Compute Equivalence, Model Diversity, and AutoGen

6.1. Multi-Agent vs. Single Agent Under the Same Token Budget

The fair comparison is not “5 agents vs. 1 agent” but rather “5 agents with X tokens vs. 1 agent with $5X$ tokens” [3]. We implemented Single-Small-XL: a single agent with 750 tokens and explicit chain-of-thought reasoning across four steps (supplier, finance, legal, compliance).

Over four independent runs, Single-Small-XL and Multi-Small converge to nearly identical means ($79.0\% \pm 3.2$ pp vs. $79.5\% \pm 7.1$ pp ARR). The 0.5 pp difference falls well within both standard errors, confirming

that with `llama-3.1-8b-instant` neither expanding the context window nor adding specialized agents produces a consistent advantage — consistent with the compute equivalence hypothesis. The expanded context improves ARR on average but shows high variability in HDA ($36.5\% \pm 13.1$ pp vs. $17.3\% \pm 8.5$ pp for Multi-Small): consolidating everything into a single prompt helps resolve more cases autonomously, but its effect on escalation detection is unstable across runs.

Li et al. [13] found that adding agents does improve performance when the base model has sufficient capacity — a result consistent with the 8B–70B threshold identified here.

6.2. Model Diversity

Table 9: Model diversity configurations with 5 agents.

Configuration	Models	ARR	HDA	DER	PT	Interpretation
Multi-Small	8B $\times$ 5	$79.5\% \pm 7.1$	17.3%	27.5%	1.74s	High variance — indistinguishable from XL
Multi-Mixed	8B $\times$ 4 + 70B $\times$ 1	$98.7\% \pm 0.9$	5.1%	27%	1.69s	Powerful decision-maker — best ARR/cost
Multi-Large	70B $\times$ 5	98%	0%	26%	2.40s	2.7 $\times$ more expensive than Multi-Mixed

The largest jump in the experiment occurs between Multi-Small ($79.5\% \pm 7.1$ pp) and Multi-Mixed (98.7%): +19.2 pp with the same agents and the same token budget. The only difference is that Multi-Mixed replaces the 8B Compliance agent with a 70B model. Model diversity reduces error correlation [33, 34], and the data show this clearly. Multi-Large achieves similar results to Multi-Mixed but costs 2.7 $\times$ more. The findings are consistent with the bottleneck having been the decision-maker, not the auxiliary agents.

6.3. Comparison with AutoGen 0.7.5

We implemented the same five-agent architecture using AutoGen 0.7.5 [3] (default RoundRobin) on the same 50 cases. ARR = 100% but DER = 76%: the system resolves everything autonomously but gets three out of four decisions wrong. HDA = 0% compared to 30.8% for CrewAI.

These results characterize AutoGen’s default behavior only — no effort was made to optimize its configuration, so this is not a fair maximum-performance comparison between frameworks. What the results do suggest is that the HiL mechanism requires explicit design regardless of the framework: it does not emerge from default settings. See Appendix G for details.

7. Ablation Study

To estimate which component contributes what value to the system [40], we ran three experiments using Multi-Small as the base system. Removing Legal (A1) raises ARR by 20 pp but DER jumps to 86%: the system approves almost everything without contractual criteria, achieving high autonomy while getting most decisions wrong. Removing Compliance (A2) produces the highest HDA (53.8%), likely because Legal integrates more context when left as the sole decision-maker. But DER = 74% makes this configuration unviable for production. A3 is the clearest result: removing both Legal and Procurement raises ARR while HDA drops to 0%. Specialization contributes those 17.3 pp of HDA at a cost of approximately 8 pp in ARR relative to the mean.

Table 10: Ablation study results. McNemar significance: * $p < 0.05$, n.s. not significant. Multi-Small baseline reports mean $\pm \sigma$ over 4 replications.

Variant	ARR	HDA	DER	Sig.	Note
Multi-Small (base)	79.5% $\pm$ 7.1	17.3%	27.5%	—	Full system
A1: No Legal	88%	38.5%	86%	*	ARR +8 pp, DER spikes
A2: No Compliance	74%	53.8%	74%	n.s.	Max HDA, unacceptable DER
A3: No Legal+Procurement	88%	0%	30%	*	High ARR, HDA eliminated
Multi-Mixed (ref.)	98.7%	5.1%	27%	**	Diversity resolves ARR

7.1. DAG Structure and Information Bottlenecks

The linear DAG topology introduces informational bottlenecks between specialized agents. Finance acts as a bridge node between operational validation (Procurement) and regulatory decision-making (Legal/Compliance): its central position means that any disruption at that point affects the entire information chain. This structural analysis, though derived from a topology that is fixed by design, is consistent with the ablation findings: removing intermediate agents alters the metrics in a non-linear fashion [41, 42].

7.2. Post-hoc Quantification of Contribution and Fragmentation

The ablation results allow us to go one step further: not just observing what happens when an agent is removed, but quantifying it. Two descriptive indicators are defined for this purpose: IFC and RCD.

Information Fragmentation Cost (IFC).. For exploratory and descriptive purposes, we define IFC as the relative ARR degradation between Simple and Complex cases:

$$\text{IFC} = 1 - \frac{\text{ARR}_{\text{Complex}}}{\text{ARR}_{\text{Simple}}} \quad (1)$$

For Multi-Small: $\text{IFC} = 0.123$ (12.3% degradation on complex cases). For Multi-Mixed: $\text{IFC} = 0$. When the decision-maker has 70B capacity, complexity produces no detectable degradation. This indicator is ad hoc and has not been externally validated; it is presented as a descriptive tool to quantify the observed effect, not as an independent methodological contribution.

Role Contribution Delta (RCD).. Measures how much each role contributes to the system’s overall HDA, computed as the performance difference after ablation:

$$\text{RCD}(\text{role}) = \text{HDA}_{\text{full}} - \text{HDA}_{\text{without_role}} \quad (2)$$

Table 11: RCD by role. Negative values indicate that removing the role improves HDA by reducing fragmentation, but at the cost of regulatory precision (DER).

Role removed	HDA without	HDA full	RCD (pp)	DER without
No Legal (A1)	38.5%	30.8%	−7.7	86%
No Compliance (A2)	53.8%	30.8%	−23.0	74%
No Legal+Proc (A3)	0%	30.8%	+30.8	30%

The negative RCD for Legal and Compliance is perhaps the most striking result of the ablation: their absence improves HDA by reducing fragmentation, but the regulatory cost is unacceptable ($\text{DER} = 74\text{--}86\%$). This quantifies the underlying problem: specialization provides local precision but fragments the context the decision-maker needs to detect systemic risk.

8. Cross-Process Validation and BPI Challenge 2019

8.1. *Signal Explicitness Gradient*

Replication across four additional processes (incident management, onboarding, compliance check, financial analysis, 20 cases each) revealed an unanticipated pattern: HDA depends not only on the architecture but on the semantic nature of the domain’s risk signals.

The pattern is fairly consistent. When risk signals are explicit and binary (incident management: risk is either present or absent), both systems reach $HDA = 100\%$. When signals are semantically complex (financial analysis, where budget overrun ratios must be interpreted in a strategic context), Single-Small drops to 0% and Multi-Mixed is the only system that detects them. Procurement sits at the hardest end of the spectrum: implicit regulatory signals (GDPR, AESA, CE certification) that require domain-specific normative knowledge.

This leads to four signal categories: **EB** (Explicit Binary), **MX** (Mixed), **IR** (Implicit Regulatory), and **SS** (Strategic Semantic). The validity of this taxonomy was verified with 2 human annotators and 4 independent LLMs (Claude Sonnet 4.6, GPT-4.5, Gemini Flash, Mistral 7B), yielding $\kappa = 0.652$ (range: [0.477–0.878], $n = 23$ signals [43, 44]). This constitutes substantial agreement. Disagreements concentrate at the MX/IR boundary, which is consistent with the semantic ambiguity of that distinction in Purchase-to-Pay contracts.

Table 12: Cross-process comparison. RPA always HDA = 0% (never escalates). RPA = RPA-Baseline; S-S = Single-Small; M-M = Multi-Mixed.

Process	Signal type	ARR			HDA		
		RPA	S-S	M-M	RPA	S-S	M-M
Procurement (50)	Implicit regulatory	100%	92%	98.7%	0%	10.3%	5.1%
Incidents (20)	Explicit binary	100%	40%	45%	0%	100%	100%
Onboarding (20)	Mixed (PEP, AML)	100%	75%	70%	0%	83.3%	100%
Compliance (20)	Mixed (REACH, OFAC)	100%	70%	65%	0%	83.3%	100%
Financial (20)	Strategic semantic	100%	100%	70%	0%	0%	100%
Target		–	>70%	>70%	–	>90%	>90%

The most illustrative case of the gradient is incident management. When signals are explicit and binary, both systems reach the HDA > 90% target without any additional tuning — something that does not occur in any other evaluated domain (Figure 4).



Figure 4: ARR and HDA metrics for incident management (explicit binary signals). Single-Small and Multi-Mixed both reach HDA = 100% when signals are directly readable.

8.2. Validation on BPI Challenge 2019

We sampled 50 cases from the BPI Challenge 2019 [1], stratified across three complexity levels, maintaining the same distribution as the main experiment. To ensure a fair comparison, the APIs were reparametrized with real data from the dataset: financial thresholds were computed from actual percentiles (P75 = 2,109 EUR, max = 8,833,459 EUR) and compliance rules were derived from the event log categories rather than artificial assumptions. The HiL escalation criterion is also objective and reproducible: a case is labeled `requires_hil=True` if it meets at least one of four conditions, all derived directly from the event log: Consignment category, Framework Order, amount exceeding 100,000 EUR, or presence of cancellation activities.

Table 13: BPI Challenge 2019 vs. synthetic dataset comparison. The fragmentation patterns replicate on real data. Synthetic values for Single-Small-XL and Multi-Small report the mean over four replications.

Model	ARR		HDA		DER	
	BPI	Synthetic	BPI	Synthetic	BPI	Synthetic
RPA-Baseline	100%	100%	0.0%	0.0%	22%	34%
Single-Small	84%	76%	45.5%	30.8%	16%	22%
Multi-Small	92%	79.5% $\pm$ 7.1	9.1%	17.3% $\pm$ 8.5	34%	27.5%
Multi-Mixed	94%	99%	18.2%	5.1%	18%	27%

Three main findings replicate on real data. First, the single agent contin-

ues to detect risk better than the multi-agent system: Single-Small achieves $HDA = 45.5\%$ vs. 9.1% for Multi-Small. The observed pattern is consistent with a context-fragmentation explanation and was also present in the real event log of a multinational company. Second, Multi-Small makes more decision errors ($DER = 34\%$ vs. 16% for Single-Small), reinforcing that multi-agent architecture with small models not only detects risk less reliably but also decides less accurately. Third, both Single-Small and Multi-Small resolve a comparable share of cases autonomously on BPI (84% vs. 92%), consistent with the compute equivalence pattern observed in the synthetic experiment — neither architecture produces a decisive advantage with 8B models.

Single-Small detects more risk in BPI (45.5%) than in the synthetic dataset (30.8%). The most plausible explanation is that BPI 2019 signals are more direct — a Consignment order or an amount above 100,000 EUR is straightforward to read. Detecting a GDPR violation or an AESA certification issue requires domain-specific normative knowledge that an 8B model does not reliably possess.

9. Discussion

9.1. Main Findings

One pattern worth examining — though the small number of HiL cases ($n = 13$) limits any strong interpretation — is that Multi-Mixed achieves $HDA = 5.1\%$, while Single-Small reaches $HDA = 10.3\%$ on average (and 30.8% in the first replication). One plausible explanation, consistent with the architecture, is the context fragmentation mechanism illustrated in Figure 5. However, given the wide confidence intervals reported in Table 6, this should be treated as a hypothesis rather than a confirmed finding.

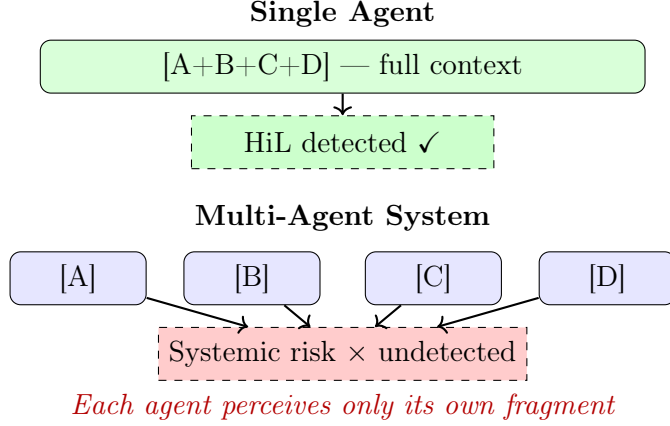


Figure 5: Context fragmentation mechanism. The single agent integrates all risk factors into one decision; in the multi-agent architecture each agent perceives only its fragment and cross-cutting risk goes undetected.

The compute equivalence pattern is also clear: over four replications, Single-Small-XL and Multi-Small converge to nearly identical ARR means (79.0% vs. 79.5%), with overlapping standard errors. With 8B models, neither expanding the context window nor adding specialized agents produces a consistent advantage. The capacity threshold appears to lie between 8B and 70B for this type of task [13].

Model diversity is the highest-impact change in the experiment: +19.2 pp in ARR simply by replacing the 8B decision-maker with a 70B model.

9.2. Connection to Organizational Theories

Three classical theories offer a lens for interpreting these patterns — though the connections are interpretive rather than empirically derived from the data.

Simon’s *bounded rationality* [45] is perhaps the most relevant analogy. Each specialized agent acts rationally within its partial context, but the collective system may fail to detect risks requiring cross-cutting integration. The ablation data are consistent with this: removing Legal and Procurement drops HDA to 0% (A3), while removing only Compliance raises it to 53.8%. This pattern suggests that intermediation structure matters, though the small n limits causal inference.

The ablation results are also consistent with Ricardo’s comparative advantage [46]: Legal detects contractual risks better than any other agent,

and Procurement handles supplier risks. The 17.3 pp of HDA lost when both are removed suggests specialization has value. The cost, however, is that no single agent holds the full risk picture — a pattern Olson [47] would describe as a collective action problem, though we apply this analogy cautiously given the exploratory nature of the HDA data.

Finance’s position in the DAG resembles what Granovetter [42] calls a weak tie: it bridges operational and regulatory decision-making. The RCD of Compliance (−23 pp) is consistent with this — removing it reduces fragmentation but raises DER to 74%, an unacceptable trade-off in practice. These theoretical parallels are offered as interpretive scaffolding, not as empirically tested claims.

9.3. Limitations

The most relevant limitation is that no configuration reached $\text{HDA} > 90\%$. The maximum in production-ready variants was 30.8%. The 10 undetected HiL cases correspond mostly to domain-specific regulations (GDPR, AESA, CE certification) that the classifier’s generic features do not capture.

The DAG is sequential and allows no iteration: if one agent makes an error, the subsequent agents cannot correct it. The 8B model is unreliable across replications ($\sigma = 11.3$ pp), which makes Single-Small results difficult to interpret reliably. On the statistical side, 13 HiL cases are too few to draw strong conclusions about HDA; the Bayes factor $\text{BF}_{10} = 0.41$ indicates the evidence is anecdotal. HDA-related analyses should be treated as exploratory given the small number of positive cases, which limits the statistical power of any between-variant comparison on this metric. Finally, the APIs, while reparametrized with real BPI 2019 data, do not behave identically to a real ERP or CRM system in production. The 50 BPI cases were selected using domain-specific stratification criteria to ensure coverage of all HiL signal types defined in the taxonomy (Consignment, Framework Order, high-value transactions, and cancellation activities). A post-hoc exploration of random sampling from the full event log revealed that unconstrained sampling produces a HiL distribution dominated by cancellation signals (>99% of positive cases), which confirms that procurement risk validation requires deliberate stratification rather than generic random sampling.

9.4. Ethical Considerations

Three ethical issues should be addressed before any real-world deployment.

The first is bias. LLMs can inherit implicit preferences from their training data — for example, favoring suppliers from certain countries without anyone having explicitly programmed that behavior. This requires periodic auditing of automated decisions.

The second is transparency. The system generates textual justifications (SAT > 3.5 in Multi-Mixed), but does not expose how it reached that decision internally. In regulated environments that is not sufficient; XAI techniques such as LIME or SHAP would need to be integrated over the final decisions.

The third is accountability. With a maximum HDA of 30.8%, a significant share of risk cases are resolved without human oversight. In real production, fixed escalation thresholds would be required — for example, any amount above 50,000 EUR or any case flagged as GDPR or AESA should always go to human review, regardless of the model’s decision.

10. Conclusions and Future Work

This work started from a concrete question: can a multi-agent system resolve in seconds what procurement today takes days to process? Technically, yes. What we did not expect is that adding more specialized agents could actually worsen risk detection.

Multi-Mixed — 8B auxiliaries, 70B decision-maker — reaches $\text{ARR} = 98.7\%$ with variability of ± 0.9 pp, processes requests in 1.74 s, and carries an estimated cost of \$0.0065 per case. But risk detection is the weak point: $\text{HDA} = 5.1\%$ in Multi-Mixed, while Single-Small achieves $\text{HDA} = 10.3\%$ on average. These figures are based on $n = 13$ positive HiL cases and should be read as preliminary evidence, not definitive results — no HDA comparison reaches statistical significance ($\text{BF}_{10} = 0.41$). Still, the pattern is consistent across replications and warrants attention: a system can autonomously resolve 98.7% of cases and still miss the ones that matter most.

Architecture and information flow matter far more than the processing capacity of each individual agent. With 8B models, neither expanding the context window nor adding specialized agents produces a consistent ARR advantage — Single-Small-XL and Multi-Small converge to the same mean over four independent runs (79.0% vs. 79.5%). Beyond ARR, specialization introduces a trade-off with risk detection: specialists become blind within their own silos, unable to pass along the context that gives meaning to what they do.

The findings suggest that the path toward higher HDA likely does not run through better models or more agents alone. One possible direction is a role whose sole purpose is not to process invoices or review regulations, but to examine the overall flow and identify the pieces that do not fit together across agents — a cross-process synthesizer: Requester $\rightarrow$ Procurement $\rightarrow$ Finance $\rightarrow$ Synthesizer $\rightarrow$ Legal $\rightarrow$ Compliance, integrating the partial outputs before the escalation decision and preserving specialization without succumbing to the silos identified here. This hypothesis requires empirical validation with more advanced architectures and additional domains.

Several lines of work remain open. The most immediate is extending the Multi-Mixed-HiL-Clf classifier with domain-specific regulatory factors (GDPR, AESA, CE certification) that are currently not captured. It also makes sense to connect the system to real ERP/CRM APIs and evaluate whether the signal explicitness gradient holds in other domains — HR, customer service, supply chain.

One final consideration: next-generation models [32] will likely shift the thresholds identified here (8B–70B). When that happens, the trade-off analysis will remain relevant. Only the location of the boundary will change.

Multi-agent architecture does not outperform the single agent by default. It does so when the underlying model has sufficient capacity to benefit from specialization — and even then, the evidence here suggests it does not guarantee better systemic risk detection. Multi-Mixed reaches $\text{ARR} = 98.7\%$ but $\text{HDA} = 5.1\%$ across only $n = 13$ positive cases, so this gap should be interpreted cautiously. What the data do support more robustly is that information flow between agents matters — redesigning it is a more promising direction than simply adding models or agents.

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References

- [1] B. van Dongen, BPI Challenge 2019, cC BY 4.0. Real-world Purchase-to-Pay event log from a multinational company. 251,734 cases. (2019).

doi:10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1.

- [2] S. Aguirre, C. Rodriguez, Automation of robotic process automation (RPA), Deloitte Insights (2017).
- [3] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, et al., AutoGen: Enabling next-gen LLM applications via multi-agent conversation framework, arXiv preprint arXiv:2308.08155 (2023).
- [4] CrewAI Inc., CrewAI: Framework for orchestrating role-playing, autonomous AI agents, <https://github.com/crewAIInc/crewAI> (2024).
- [5] LangChain Inc., LangGraph: Building stateful multi-agent applications, <https://github.com/langchain-ai/langgraph> (2024).
- [6] C. Qian, X. Cong, C. Yang, W. Chen, Y. Su, J. Xu, Z. Liu, M. Sun, Communicative agents for software development, arXiv preprint arXiv:2307.07924 (2023).
- [7] S. Hong, M. Zhuge, J. Chen, X. Zheng, Y. Cheng, C. Zhang, J. Wang, Z. Wang, S. K. S. Yau, Z. Lin, L. Zhou, C. Ran, L. Xiao, C. Wu, J. Schmidhuber, MetaGPT: Meta programming for a multi-agent collaborative framework, in: The Twelfth International Conference on Learning Representations (ICLR), 2024.
URL <https://openreview.net/forum?id=VtmBAGCN7o>
- [8] W. Chen, Y. Su, J. Zuo, C. Yang, C. Yuan, C.-M. Chan, H. Yu, Y. Lu, Y.-H. Hung, C. Qian, Y. Qin, X. Cong, R. Xie, Z. Liu, M. Sun, J. Zhou, AgentVerse: Facilitating multi-agent collaboration and exploring emergent behaviors, in: The Twelfth International Conference on Learning Representations (ICLR), 2024.
URL <https://arxiv.org/abs/2308.10848>
- [9] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, J.-R. Wen, A survey on large language model based autonomous agents, *Frontiers of Computer Science* 18 (6) (2024) 186345. doi:10.1007/s11704-024-40231-1.

- [10] Y. Talebirad, A. Nadiri, Multi-agent collaboration: Harnessing the power of intelligent LLM agents, arXiv preprint arXiv:2306.03314 (2023).
- [11] J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, M. S. Bernstein, Generative agents: Interactive simulacra of human behavior, arXiv preprint arXiv:2304.03442 (2023).
- [12] G. Li, H. A. A. K. Hammoud, H. Itani, D. Khizbullin, B. Ghanem, CAMEL: Communicative agents for “mind” exploration of large scale language model society, arXiv preprint arXiv:2303.17760 (2023).
- [13] J. Li, Q. Zhang, Y. Yu, Q. Fu, D. Ye, More agents is all you need, arXiv preprint arXiv:2402.05120 (2024).
- [14] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, arXiv preprint arXiv:2201.11903 (2022).
- [15] S. Yao, J. Zhao, D. Yu, I. Shafran, K. Narasimhan, Y. Cao, ReAct: Synergizing reasoning and acting in language models, arXiv preprint arXiv:2210.03629 (2023).
- [16] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, ACM Computing Surveys 55 (12) (2023) 248:1–248:38.
- [17] UiPath, UiPath platform: Robotic process automation, <https://www.uipath.com> (2023).
- [18] Automation Anywhere, Automation anywhere: Intelligent automation platform, <https://www.automationanywhere.com> (2023).
- [19] A. Drouin, M. Gasse, M. Caccia, I. H. Laradji, M. D. Verme, T. Marty, L. Boisvert, M. Thakkar, Q. Cappart, D. Vazquez, N. Chapados, A. Lacoste, WorkArena: How capable are web agents at solving common knowledge work tasks?, in: Proceedings of the 41st International Conference on Machine Learning, 2024.
- [20] S. Qiao, R. Fang, N. Zhang, Y. Luo, W. Zhou, Y. E. Jiang, C. Lv, H. Chen, TaskBench: Benchmarking large language models for task automation, arXiv preprint arXiv:2311.18760 (2023).

- [21] Y. Qin, S. Liang, Y. Ye, K. Zhu, L. Yan, Y. Lu, Y. Lin, X. Cong, X. Tang, B. Qian, et al., ToolLLM: Facilitating large language models to master 16000+ real-world APIs, arXiv preprint arXiv:2307.16789 (2023).
- [22] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, T. Scialom, Toolformer: Language models can teach themselves to use tools, in: Advances in Neural Information Processing Systems (NeurIPS), 2023.
- [23] S. G. Patil, T. Zhang, X. Wang, J. E. Gonzalez, Gorilla: Large language model connected with massive APIs, arXiv preprint arXiv:2305.15334 (2023).
- [24] W. M. P. van der Aalst, A. Adriansyah, A. K. A. de Medeiros, et al., Process mining manifesto (2011) 169–194.
- [25] W. M. P. van der Aalst, Process mining: Overview and opportunities, ACM Transactions on Management Information Systems 3 (2) (2012) 7:1–7:17.
- [26] B. F. van Dongen, BPI challenge 2011, in: Business Process Intelligence Challenge, 2011.
- [27] European Parliament and Council, Regulation (EU) 2016/679 — general data protection regulation (GDPR), Official Journal of the European Union, L 119/1 (2016).
- [28] D. Mosqueira-Rey, E. Hernández-Pereira, D. Alonso-Ríos, J. Bobes-Bascarán, A. Fernández-Leal, Human-in-the-loop machine learning: A state of the art, Artificial Intelligence Review 56 (4) (2023) 3005–3054. doi:10.1007/s10462-022-10246-w.
- [29] Meta AI, The llama 3 herd of models, arXiv preprint arXiv:2407.21783 (2024).
- [30] Groq Inc., Groq: Fast AI inference, <https://groq.com> (2024).
- [31] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, et al., Language models are few-shot learners (2020).
- [32] OpenAI, GPT-4 technical report, arXiv preprint arXiv:2303.08774 (2023).

- [33] T. G. Dietterich, Ensemble methods in machine learning (2000) 1–15.
- [34] H. M. Gomes, A. Bifet, J. Read, J. P. Barddal, F. Enembreck, B. Pfharinger, G. Holmes, T. Abdessalem, Adaptive random forests for evolving data stream classification, *Machine Learning* 106 (9) (2017) 1469–1495.
- [35] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, K. Narasimhan, SWE-bench: Can language models resolve real-world GitHub issues?, *arXiv preprint arXiv:2310.06770* (2024).
- [36] G. Mialon, C. Fourrier, C. Swift, T. Wolf, Y. LeCun, T. Scialom, GAIA: A benchmark for general AI assistants, *arXiv preprint arXiv:2311.12983* (2023).
- [37] X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang, et al., AgentBench: Evaluating LLMs as agents, *arXiv preprint arXiv:2308.03688* (2023).
- [38] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, et al., Judging LLM-as-a-Judge with MT-Bench and chatbot arena, *arXiv preprint arXiv:2306.05685* (2023).
- [39] S. Ouyang, J. M. Zhang, M. Harman, M. Wang, LLM is like a box of chocolates: The non-determinism of ChatGPT in code generation, *arXiv preprint arXiv:2308.02828* (2023).
- [40] S. Bubeck, et al., Sparks of artificial general intelligence: Early experiments with GPT-4, *arXiv preprint arXiv:2303.12528* (2023).
- [41] L. C. Freeman, A set of measures of centrality based on betweenness, *Sociometry* 40 (1) (1977) 35–41.
- [42] M. S. Granovetter, The strength of weak ties, *American Journal of Sociology* 78 (6) (1973) 1360–1380.
- [43] J. Cohen, A coefficient of agreement for nominal scales, *Educational and Psychological Measurement* 20 (1) (1960) 37–46.
- [44] J. R. Landis, G. G. Koch, The measurement of observer agreement for categorical data, *Biometrics* 33 (1) (1977) 159–174.

- [45] H. A. Simon, A behavioral model of rational choice, *The Quarterly Journal of Economics* 69 (1) (1955) 99–118.
- [46] D. Ricardo, *On the Principles of Political Economy and Taxation*, John Murray, 1817.
- [47] M. Olson, *The Logic of Collective Action: Public Goods and the Theory of Groups*, Harvard University Press, 1965.
- [48] Elsevier, Generative AI policies for journals, <https://www.elsevier.com/about/policies-and-standards/generative-ai-policies-for-journals>, accessed: May 2026 (2025).

Appendix A. Agent System Prompts

The prompts for all five agents are included below. The complete set with additional parameters is available in the public repository.

Requester Agent

```
role: "Request Manager"
goal: "Register and structure purchase requests."
backstory: "You are the first point of contact. You register requests
            and prepare them for the team."
```

Finance Agent

```
role: "Finance Director"
goal: "Verify budget availability and validate the requested amount."
backstory: "You are responsible for budget control. You check the
            department's available budget and determine whether the
            requested amount is within approved limits."
```

Procurement Agent

```
role: "Procurement Manager"
goal: "Validate the supplier and verify their approval status."
backstory: "You are a specialist in supplier management. You verify
            that the supplier is approved, has no conflict-of-interest
            alerts, and meets company requirements."
```

Legal Agent

```
role: "Senior Legal Advisor"
goal: "Review contractual compliance and detect legal risks."
backstory: "You are familiar with GDPR, AESA, CE, and PRL regulations.
            You analyze risk clauses in contracts and issue a legal
            recommendation."
```

Compliance Agent

```
role: "Compliance Director"
goal: "Verify regulatory compliance and escalate when risk exceeds
       established thresholds."
backstory: "You are the final decision-maker. You have authority to
            approve, reject, or escalate to the responsible human."
```

Appendix B. Reproducibility Parameters

- Framework: CrewAI, Python 3.12, Windows 10, Jupyter Lab
- Auxiliary LLM: llama-3.1-8b-instant via Groq API
- Decision LLM (Mixed/Large): llama-3.3-70b-versatile via Groq API
- Temperature: 0.0 (with the caveat documented in Sec. 4)
- Anti-rate-limit pause: 4–8 s between cases
- Code and datasets: <https://github.com/Edurne0781/multiagent-procurement-llm>

The BPI Challenge 2019 is available at DOI: 10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1 (CC BY 4.0).

Appendix C. BPI Challenge 2019 Annotation Protocol

A case is labeled `requires_hil=True` if it meets at least one of the following criteria derived directly from the event log: (a) category `Consignment`; (b) document type `Framework order`; (c) amount exceeding 100,000 EUR; or (d) presence of cancellation activities (`Cancel Invoice Receipt` or `Cancel Goods Receipt`).

Appendix D. Inter-Annotator Kappa — Risk Signal Taxonomy

Taxonomy validation (EB, MX, IR, SS) was carried out with 2 human annotators and 4 independent LLMs. $\kappa = 0.652$ (range: [0.477–0.878]). Disagreements concentrate at the MX/IR boundary. The detailed pairwise annotator table is available in the repository.

Appendix E. Justification Quality: LLM-as-Judge

The cost comparison in Figure 3 shows that Multi-Small and Single-Small-XL share the exact same token budget (\$0.0045/case), yet Single-Small-XL achieves an ARR 16 pp higher. That gap is the compute equivalence hypothesis expressed in numbers. The SAT scores below reflect justification quality for the main variants.

Table E.14: LLM-as-Judge evaluation of justification quality (1–5 scale, $n = 15$ cases). SAT = mean(Clarity, Completeness, Actionability). Target: SAT > 3.5.

Variant	Clarity	Completeness	Actionability	SAT
RPA-Baseline	3.67	3.20	2.40	3.09 ± 0.67
Single-Small	4.20	3.60	4.00	3.93 ± 0.69
Multi-Small	4.07	3.86	4.14	4.02 ± 0.55
Multi-Mixed	4.27	4.53	4.27	4.36 ± 0.27

The justification quality results are consistent with the ARR findings: Multi-Mixed not only resolves more cases but also explains its decisions more clearly. The actionability score of RPA-Baseline (2.40) reflects its core limitation precisely — it approves or rejects without giving the recipient any useful context.

Appendix F. Qualitative Error Analysis

Case C40 (GDPR + foreign cloud supplier).. Multi-Mixed approves it: supplier is registered, budget is available. Single-Small escalates it in replication 1 by detecting the combination of foreign supplier and personal data. The GDPR risk is not captured as an explicit factor in the mock APIs.

Case C43 (Conflict of interest + high amount).. This case was correctly detected. Multi-Mixed-HiL-Clf escalates it: the supplier has a conflict-of-interest alert and the amount exceeds the board approval threshold. This is the profile the classifier handles well: quantifiable, combined signals.

Appendix G. Detailed AutoGen vs. CrewAI Comparison

AutoGen 0.7.5 with default RoundRobin on the same 50 cases: ARR = 100%, HDA = 0%, DER = 76%, PT = 6.55 s. Processing time is nearly four times that of CrewAI (Multi-Mixed: 1.69 s). These results characterize default behavior, not the maximum achievable with AutoGen. The relevant finding: HDA = 0% in both frameworks under basic configuration, which supports the conclusion that the HiL mechanism requires explicit design.

Appendix H. Multi-Mixed-HiL-Clf Classifier Pseudocode

```

1 def hil_classifier(case):
2     # Stage 1: Uncertainty Evaluator
3     factors = extract_risk_factors(case)
4     confidence = evaluate_confidence(factors) # "high"|"low"
5     if confidence == "low":
6         return escalate_to_hil(case, summary=factors)
7     # Stage 2: Senior Manager
8     return senior_manager_decides(case, factors)

```

Data Availability Statement

The BPI Challenge 2019 dataset ($n = 251,734$) is publicly available under CC BY 4.0 at DOI: 10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1 [1]. The synthetic dataset (50 cases), the five secondary datasets (130 cases), the code, and the system prompts are available at: <https://github.com/Edurne0781/multiagent-procurement-llm>. The repository will be archived on Zenodo with a permanent DOI prior to publication.

Ethics Statement

This work does not involve human subjects, personal data, or animal experiments. The BPI Challenge 2019 contains anonymized industrial logs (CC BY 4.0). The inter-annotator validation involved voluntary participation with no collection of personal data. No ethical approval was required.

CRedit Author Contribution Statement

Edurne Martínez de Contrasta: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the author used generative AI tools — specifically Claude Sonnet 4.6 (Anthropic) — to assist with writing tasks, grammatical revision, and text structuring. These tools played no role in the experimental design, data generation, statistical analysis, or interpretation of results. All content was reviewed, edited, and validated by the author, who takes full responsibility for the accuracy, integrity, and originality of the manuscript. This declaration complies with Elsevier’s policy on the use of generative AI [48].